

Charleston County Sample — Shopping Center / Retail



Traffic Impact Study

Prepared August 11, 2026 · 32.8560, -79.9750

Simple Impact Studies — sample draft

Traffic Impact Study

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EXECUTIVE SUMMARY

This Traffic Impact Study evaluates the proposed Charleston County Sample — Shopping Center / Retail within Charleston (SC) MSA, South Carolina (Charleston region (Charleston / Berkeley / Dorchester)), in accordance with SCDOT's Access and Roadside Management Standards (ARMS, 2008 edition, updated July 8, 2025) Chapter 6 and Traffic Engineering Guideline TG-21. In South Carolina the TIS is an element of the SCDOT encroachment-permit process: the study scope must be established with the District Traffic Engineer (DTE) before analysis begins, and the DTE makes the final mitigation determination.

12	680	0	1
INTERSECTIONS	PM PEAK TRIPS	AT LOS E/F	LOS DROPS

1.0 INTRODUCTION

Project name	Charleston County Sample — Shopping Center / Retail
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	32.8560°, -79.9750°
Build-out year	2027
Controlling jurisdiction	Charleston region (Charleston / Berkeley / Dorchester)
MPO / study	CHATS (Charleston Area Transportation Study), staffed by BCDCOG
Programmed projects source	SCDOT FFY 2024–2033 STIP, as amended (e-STIP)
Submittal channel	SCDOT EPPS (Encroachment Permit Processing System) via the District Permit Engineer

1.1 TIS Applicability

City of Charleston TIS Preparation Guide (June 2021): a TIS is required when a development includes ANY of — a drive-through window; more than 6 fuel dispensers; more than 10,000 sf of non-residential building coverage; a site of 5 acres or more; a driveway-spacing variance request; a restaurant over 4,000 sf GFA; or single/two-family projects totaling 45 or more units. The statewide ARMS 100 peak-hour-trip trigger applies on SCDOT facilities.

The proposed development's estimated peak-hour generation of 680 trips meets or exceeds the ARMS 100 peak-hour-trip threshold; a TIS is required for the encroachment permit.

2.0 PROJECT TRAFFIC

ARMS Chapter 6 calls for trip generation per the latest edition of the ITE Trip Generation Manual, with internal-capture, pass-by, and transit reductions justified to the DTE. This screening analysis applies public-data trip-generation rates (NHTS 2017 / SANDAG 2002 / NCHRP 716), with every rate tagged so the jurisdiction-approved ITE figure can be substituted at submittal.

SCDOT District 6 caps for Charleston-area studies: internal capture limited to 20% of the lesser of entering/exiting trips; pass-by limited to 10% of adjacent street traffic. The credits applied in this screening analysis must be checked against these caps at submittal.

Daily trips (gross)	6,800
AM peak hour	166 in / 106 out
PM peak hour	326 in / 354 out

3.0 TRAFFIC VOLUME DEVELOPMENT AND CAPACITY ANALYSIS

Scenarios per ARMS: existing year, build-out no-build, and build-out build. Capacity analysis follows the Highway Capacity Manual (HCM 6th Edition consistent) with LOS reported for all approaches and movements; coordinated signal systems are analyzed as a system.

3.1 TG-21 Level of Service Standard — LOS C

Per SCDOT Traffic Engineering Guideline TG-21 (Mitigation of Traffic Impacts), the acceptable level of service for the design (peak) hour is LOS C or better for all roadway types statewide, applied in lieu of locally preferred thresholds. Where the baseline already operates at or below LOS C, the baseline LOS must be maintained or improved; where the baseline is LOS F in a congested urban area, mitigation is at the DTE's determination. Note: this is a stricter standard than the LOS D convention applied in several neighboring states.

No.	Intersection	Exist.	No-Build	Build LOS/Delay	TG-21 (LOS C) check
1	Cosgrove Avenue & Rivers Avenue	B	B	B / 20.0s	meets LOS C
2	Rivers Avenue & Reynolds Avenue	B	B	C / 20.9s	meets LOS C
3	Cosgrove Avenue & Spruill Avenue	B	B	B / 18.8s	meets LOS C
4	Dorchester Road & Rivers Avenue	B	B	B / 18.4s	meets LOS C
5	Reynolds Avenue & Spruill Avenue	B	B	B / 16.7s	meets LOS C
6	McMillan Avenue & Spruill Avenue	B	B	B / 19.8s	meets LOS C
7	Dorchester Road & Meeting Street Road	B	B	B / 17.7s	meets LOS C
8	Rivers Avenue & McMillan Avenue	B	B	B / 19.4s	meets LOS C
9	Spruill Avenue & North Carolina Avenue	B	B	B / 17.0s	meets LOS C
10	North Hobson Avenue & Noisette Boulevard	B	B	B / 18.2s	meets LOS C
11	Cosgrove Avenue & Azalea Drive	B	B	B / 19.5s	meets LOS C
12	Near Dorchester Road	B	B	B / 14.2s	meets LOS C

4.0 FINDINGS, ACCESS MANAGEMENT, AND SIGNAL WARRANTS

1 intersection project a LOS drop and 0 operate at LOS E or F under Build conditions; mitigation alternatives (turn lanes with storage per ARMS Ch. 5 and the SCDOT Roadway Design Manual Ch. 9, signalization, or operational improvements) are developed with the DTE, who makes the final determination.

Access management: ARMS requires demonstration that the fewest driveways necessary serve the site, with sight-distance verification at each access point. Signal proposals require a signal warrant analysis per the MUTCD. Both are site-plan-dependent and are prepared at submittal.

5.0 ARMS TECHNICAL COMPLETENESS CHECKLIST

SCDOT returns incomplete studies unreviewed. The ARMS completeness checklist is tracked below against this screening draft; unmet items are produced at formal submittal.

Checklist item	Status in this draft
SC PE stamp and signature	At submittal (screening draft is unsealed)
Introduction + executive summary	Included

Checklist item	Status in this draft
Existing conditions (classification, peaks, LOS, volumes)	Included (screening level)
Trip generation / distribution / LOS analysis	Included (public-data rates, tagged)
Crash data / speeds / sight distance	At submittal (site-specific)
Mitigation w/ turn-lane storage lengths	Indicated per TG-21 table; design at submittal
Figures (vicinity, site plan, volumes, lane configs, LOS)	Volume/LOS figures included; site-plan figures at submittal
Tables (trip gen, LOS existing/background/build/mitigated)	Included (screening level)
Technical appendix (capacity reports, warrants)	Capacity worksheets appended; Synchro/warrants at submittal

This TIS is to be prepared under the direct charge of and sealed by a registered South Carolina Professional Engineer with expertise in traffic engineering (ARMS), with seal, signature across the seal face, and date per S.C. Code §40-22-270 and Regs. Ch. 49, and the firm's Certificate of Authorization. Electronic sealing is accepted in current practice.

SCREENING DISCLOSURE: this document is a screening-level draft prepared before DTE scoping. Scope, growth rates, background developments, and mitigation are established with the District Traffic Engineer; a submittal-grade analysis supersedes this draft.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work: $a=28507$, $b=-0.02$, $c=-0.123$) on the travel time t to the signal.

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Cosgrove Avenue & Rivers Avenue	0.11	0.3	253	23.1%
Rivers Avenue & Reynolds Avenue	0.17	0.4	189	17.3%
Cosgrove Avenue & Spruill Avenue	0.23	0.6	139	12.7%
Dorchester Road & Rivers Avenue	0.24	0.6	134	12.2%
Reynolds Avenue & Spruill Avenue	0.28	0.7	111	10.1%
McMillan Avenue & Spruill Avenue	0.32	0.8	88	8.0%
Dorchester Road & Meeting Street Road	0.40	1.0	59	5.4%
Rivers Avenue & McMillan Avenue	0.43	1.0	51	4.7%
Spruill Avenue & North Carolina Avenue	0.60	1.4	22	2.0%
North Hobson Avenue & Noisette Boulevard	0.62	1.5	20	1.8%
Cosgrove Avenue & Azalea Drive	0.67	1.6	15	1.4%
Near Dorchester Road	0.71	1.7	13	1.2%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 94% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 6% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

A capacity-constrained assignment using the BPR volume-delay function $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives until the assignment is congestion-consistent. Highest post-build v/c among study signals: 0.56. Per-signal v/c , delay, LOS, and queue are in the capacity appendix. (Road-network corridor routing was not available for this region; per-signal capacity-constrained assignment was used.)

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

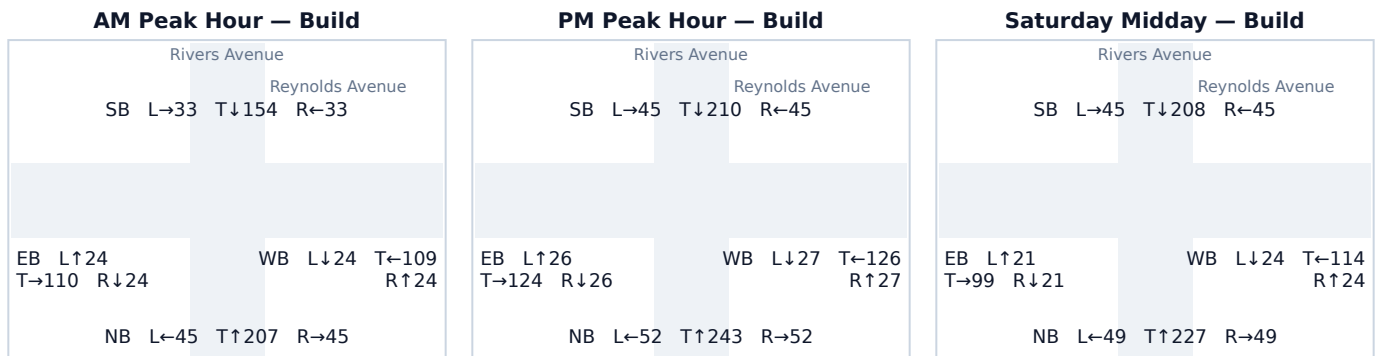
One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach HCM capacity table. Analysis follows the Highway Capacity Manual 6th Edition signalized-intersection method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 Rivers Avenue & Reynolds Avenue

Signal ID	charleston-sc-147
Location	Central Charleston (SC) · 0.17 mi from site
Added PM peak trips	189
Existing / No-Build (PM)	LOS B · 18.9 s/veh · v/c 0.45
Build (PM)	LOS C · 20.9 s/veh · v/c 0.56
Δ control delay	+2.0 s/veh (LOS change)
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	280	66	347	0.35 → 0.43	17.3 → 18.5	B → B	243
SB	201	99	300	0.25 → 0.37	16.1 → 17.6	B → B	204
EB	176	0	176	0.22 → 0.22	15.7 → 15.7	B → B	111
WB	156	24	180	0.19 → 0.22	15.4 → 15.8	B → B	114

Affected movements (PM peak project trips)

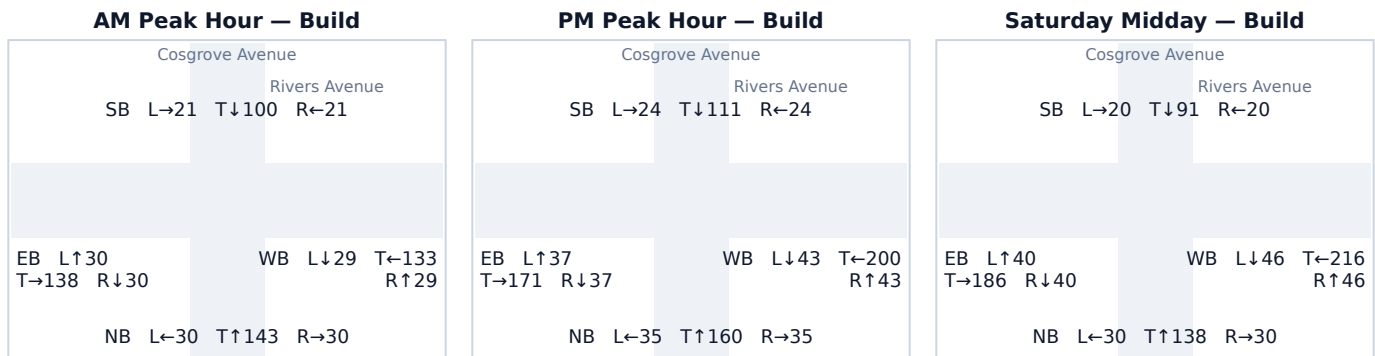
Movement	Project trips	% of project trips
SB Through	72	38%
NB Through	66	35%
SB Left	27	14%
WB Right	24	13%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 Cosgrove Avenue & Rivers Avenue

Signal ID	charleston-sc-301
Location	NW Charleston (SC) · 0.11 mi from site
Added PM peak trips	253
Existing / No-Build (PM)	LOS B · 17.6 s/veh · v/c 0.37
Build (PM)	LOS B · 20.0 s/veh · v/c 0.51
Δ control delay	+2.4 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	211	19	230	0.26 → 0.28	16.2 → 16.5	B → B	150
SB	154	5	159	0.19 → 0.20	15.4 → 15.5	B → B	99
EB	147	97	245	0.18 → 0.30	15.3 → 16.7	B → B	161
WB	155	132	286	0.19 → 0.35	15.4 → 17.4	B → B	193

Affected movements (PM peak project trips)

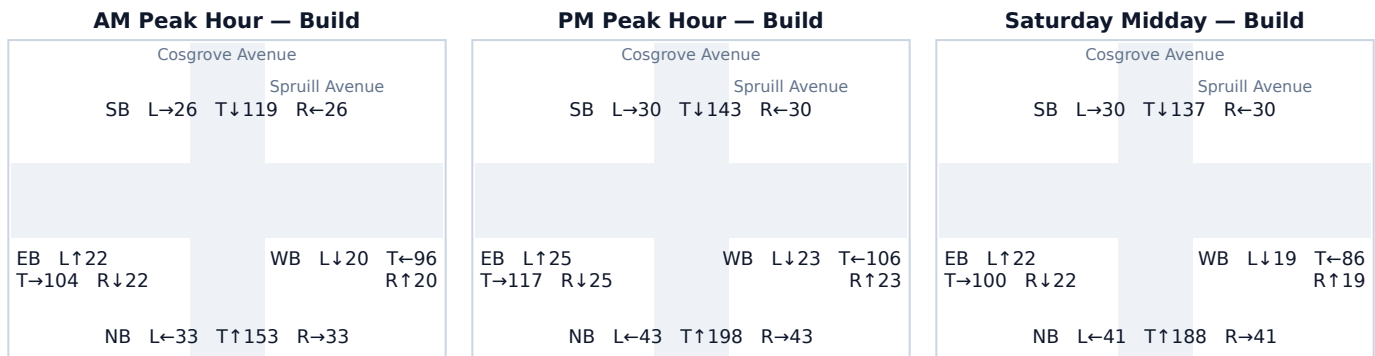
Movement	Project trips	% of project trips
WB Through	106	42%
EB Through	97	38%
WB Left	21	8%
NB Right	19	8%
WB Right	5	2%
SB Left	5	2%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 Cosgrove Avenue & Spruill Avenue

Signal ID	charleston-sc-78
Location	NW Charleston (SC) · 0.23 mi from site
Added PM peak trips	139
Existing / No-Build (PM)	LOS B · 17.6 s/veh · v/c 0.37
Build (PM)	LOS B · 18.8 s/veh · v/c 0.45
Δ control delay	+1.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	212	72	284	0.26 → 0.35	16.2 → 17.4	B → B	191
SB	154	49	203	0.19 → 0.25	15.4 → 16.1	B → B	130
EB	153	14	167	0.19 → 0.21	15.4 → 15.6	B → B	105
WB	148	4	152	0.18 → 0.19	15.3 → 15.4	B → B	94

Affected movements (PM peak project trips)

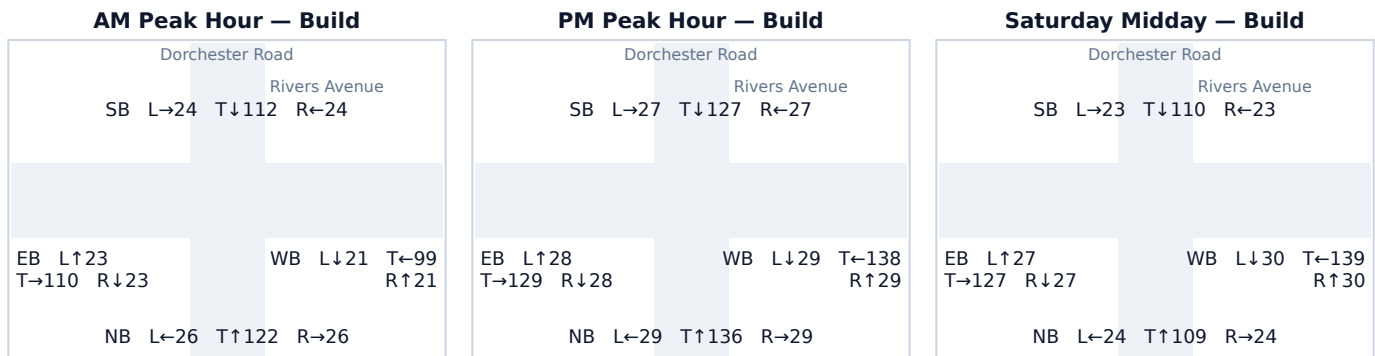
Movement	Project trips	% of project trips
NB Through	53	38%
SB Through	49	35%
NB Left	15	11%
EB Right	14	10%
NB Right	4	3%
WB Left	4	3%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 Dorchester Road & Rivers Avenue

Signal ID	charleston-sc-304
Location	NW Charleston (SC) · 0.24 mi from site
Added PM peak trips	134
Existing / No-Build (PM)	LOS B · 17.3 s/veh · v/c 0.35
Build (PM)	LOS B · 18.4 s/veh · v/c 0.42
Δ control delay	+1.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	191	3	194	0.24 → 0.24	15.9 → 16.0	B → B	123
SB	166	15	181	0.20 → 0.22	15.6 → 15.8	B → B	114
EB	138	47	185	0.17 → 0.23	15.2 → 15.8	B → B	117
WB	126	69	196	0.16 → 0.24	15.0 → 16.0	B → B	125

Affected movements (PM peak project trips)

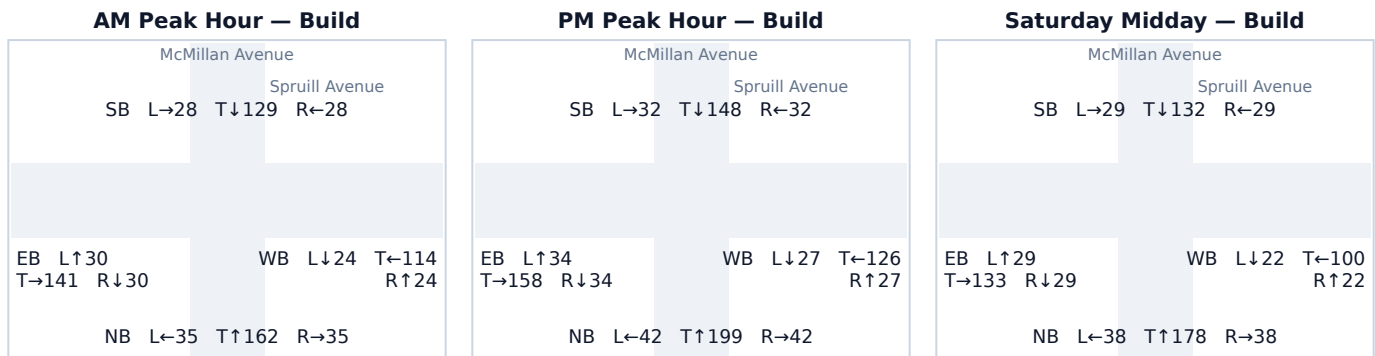
Movement	Project trips	% of project trips
WB Through	50	37%
EB Through	47	35%
WB Right	16	12%
SB Left	15	11%
WB Left	3	2%
NB Right	3	2%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 McMillan Avenue & Spruill Avenue

Signal ID	charleston-sc-77
Location	NW Charleston (SC) · 0.32 mi from site
Added PM peak trips	88
Existing / No-Build (PM)	LOS B · 18.9 s/veh · v/c 0.45
Build (PM)	LOS B · 19.8 s/veh · v/c 0.50
Δ control delay	+0.9 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	237	46	283	0.29 → 0.35	16.6 → 17.3	B → B	191
SB	184	28	212	0.23 → 0.26	15.8 → 16.2	B → B	136
EB	212	14	226	0.26 → 0.28	16.2 → 16.4	B → B	147
WB	179	0	180	0.22 → 0.22	15.8 → 15.8	B → B	113

Affected movements (PM peak project trips)

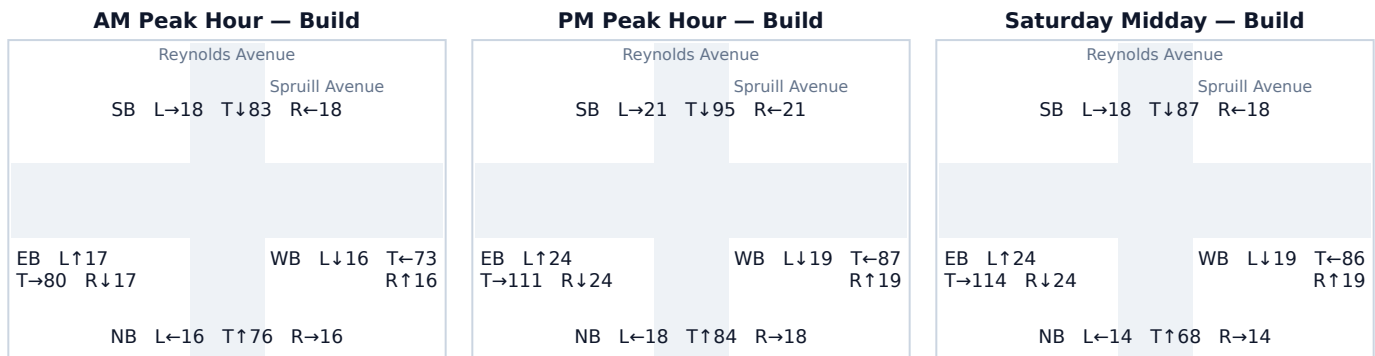
Movement	Project trips	% of project trips
NB Through	31	35%
SB Through	28	32%
NB Left	15	17%
EB Right	14	16%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 Reynolds Avenue & Spruill Avenue

Signal ID	charleston-sc-79
Location	NW Charleston (SC) · 0.28 mi from site
Added PM peak trips	111
Existing / No-Build (PM)	LOS B · 15.9 s/veh · v/c 0.24
Build (PM)	LOS B · 16.7 s/veh · v/c 0.30
Δ control delay	+0.8 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	119	0	120	0.15 → 0.15	15.0 → 15.0	B → B	73
SB	118	19	137	0.15 → 0.17	14.9 → 15.2	B → B	84
EB	101	58	159	0.12 → 0.20	14.7 → 15.5	B → B	99
WB	91	34	125	0.11 → 0.15	14.6 → 15.0	B → B	76

Affected movements (PM peak project trips)

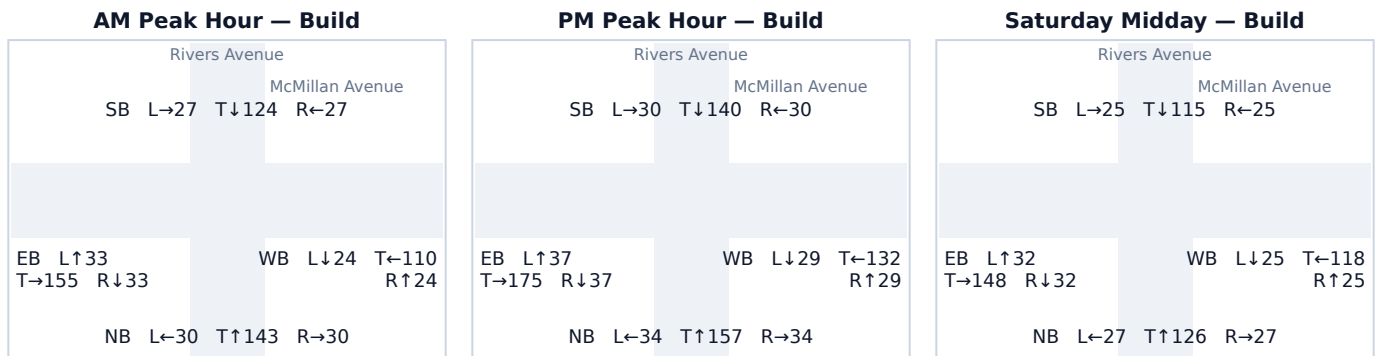
Movement	Project trips	% of project trips
EB Through	37	33%
WB Through	34	31%
EB Left	20	18%
SB Right	19	17%
EB Right	1	1%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 Rivers Avenue & McMillan Avenue

Signal ID	charleston-sc-303
Location	NW Charleston (SC) · 0.43 mi from site
Added PM peak trips	51
Existing / No-Build (PM)	LOS B · 18.9 s/veh · v/c 0.45
Build (PM)	LOS B · 19.4 s/veh · v/c 0.48
Δ control delay	+0.5 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	225	0	225	0.28 → 0.28	16.4 → 16.4	B → B	146
SB	192	7	200	0.24 → 0.25	15.9 → 16.0	B → B	127
EB	232	17	249	0.29 → 0.31	16.5 → 16.8	B → B	164
WB	164	27	190	0.20 → 0.23	15.5 → 15.9	B → B	121

Affected movements (PM peak project trips)

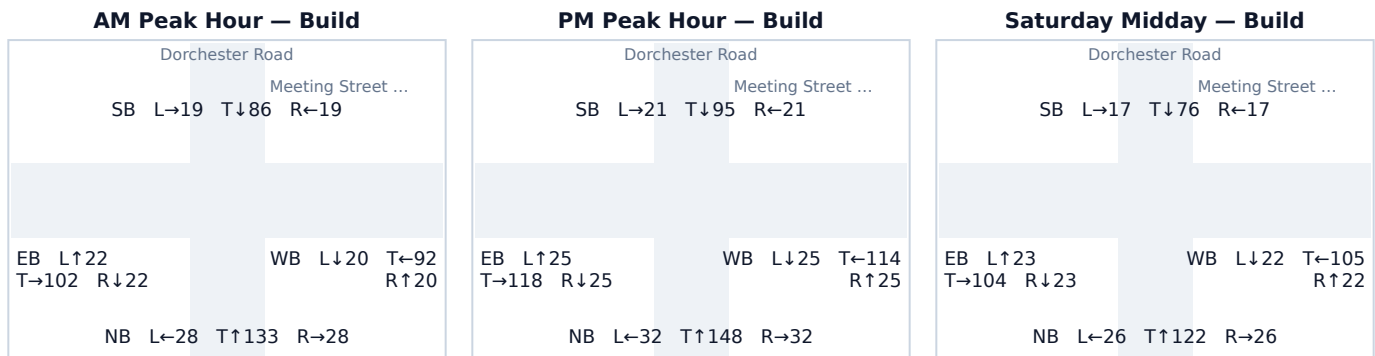
Movement	Project trips	% of project trips
WB Through	19	37%
EB Through	17	33%
WB Right	8	16%
SB Left	7	14%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.8 Dorchester Road & Meeting Street Road

Signal ID	charleston-sc-362
Location	NW Charleston (SC) · 0.40 mi from site
Added PM peak trips	59
Existing / No-Build (PM)	LOS B · 17.3 s/veh · v/c 0.35
Build (PM)	LOS B · 17.7 s/veh · v/c 0.38
Δ control delay	+0.4 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	205	7	212	0.25 → 0.26	16.1 → 16.2	B → B	136
SB	137	0	137	0.17 → 0.17	15.2 → 15.2	B → B	84
EB	146	22	168	0.18 → 0.21	15.3 → 15.6	B → B	105
WB	133	30	164	0.16 → 0.20	15.1 → 15.5	B → B	102

Affected movements (PM peak project trips)

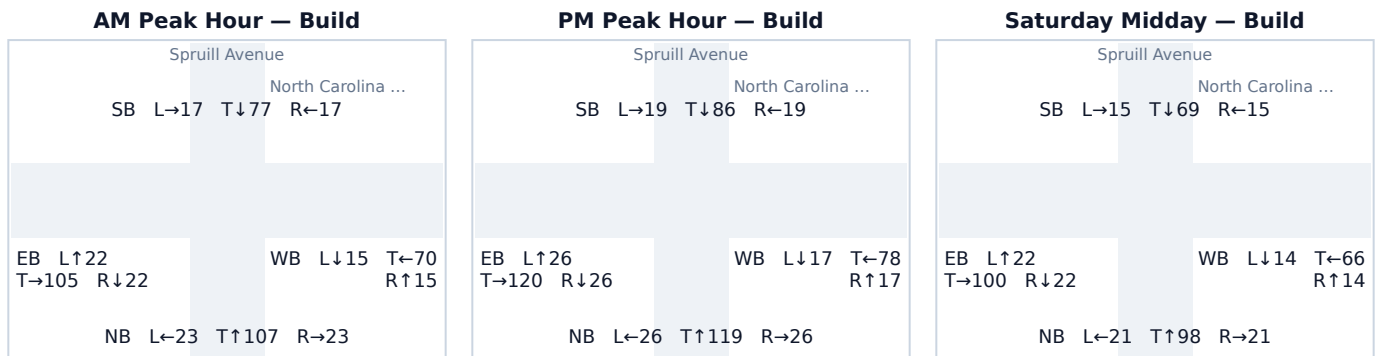
Movement	Project trips	% of project trips
WB Through	23	39%
EB Through	22	37%
WB Left	7	12%
NB Right	7	12%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.9 Spruill Avenue & North Carolina Avenue

Signal ID	charleston-sc-753
Location	Central Charleston (SC) · 0.60 mi from site
Added PM peak trips	22
Existing / No-Build (PM)	LOS B · 16.8 s/veh · v/c 0.31
Build (PM)	LOS B · 17.0 s/veh · v/c 0.32
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	167	4	171	0.21 → 0.21	15.6 → 15.6	B → B	107
SB	123	0	124	0.15 → 0.15	15.0 → 15.0	B → B	75
EB	161	12	172	0.20 → 0.21	15.5 → 15.6	B → B	108
WB	106	6	112	0.13 → 0.14	14.8 → 14.9	B → B	68

Affected movements (PM peak project trips)

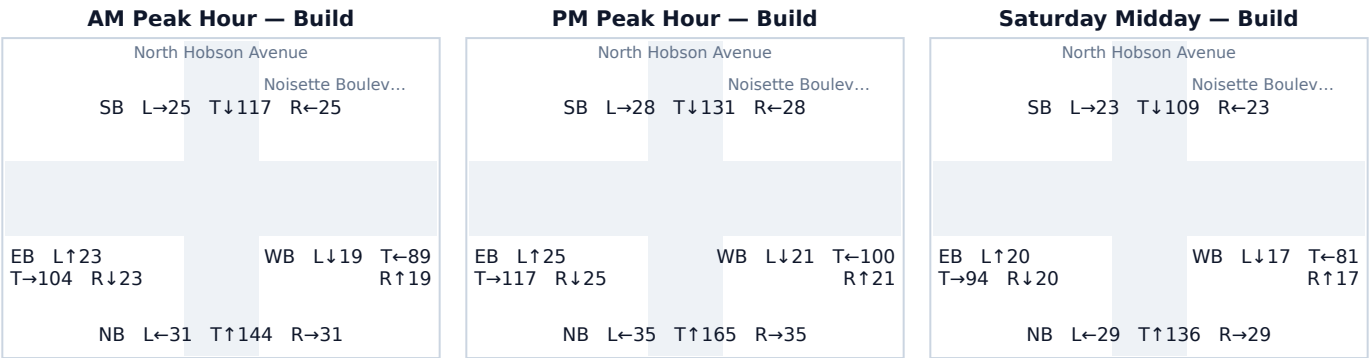
Movement	Project trips	% of project trips
EB Through	7	32%
WB Through	6	27%
EB Right	4	18%
NB Left	4	18%
EB Left	1	5%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.10 North Hobson Avenue & Noisette Boulevard

Signal ID	charleston-sc-82
Location	NW Charleston (SC) · 0.62 mi from site
Added PM peak trips	20
Existing / No-Build (PM)	LOS B · 18.0 s/veh · v/c 0.39
Build (PM)	LOS B · 18.2 s/veh · v/c 0.41
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	225	11	235	0.28 → 0.29	16.4 → 16.6	B → B	153
SB	180	7	187	0.22 → 0.23	15.8 → 15.9	B → B	119
EB	167	0	167	0.21 → 0.21	15.6 → 15.6	B → B	104
WB	139	2	142	0.17 → 0.17	15.2 → 15.2	B → B	87

Affected movements (PM peak project trips)

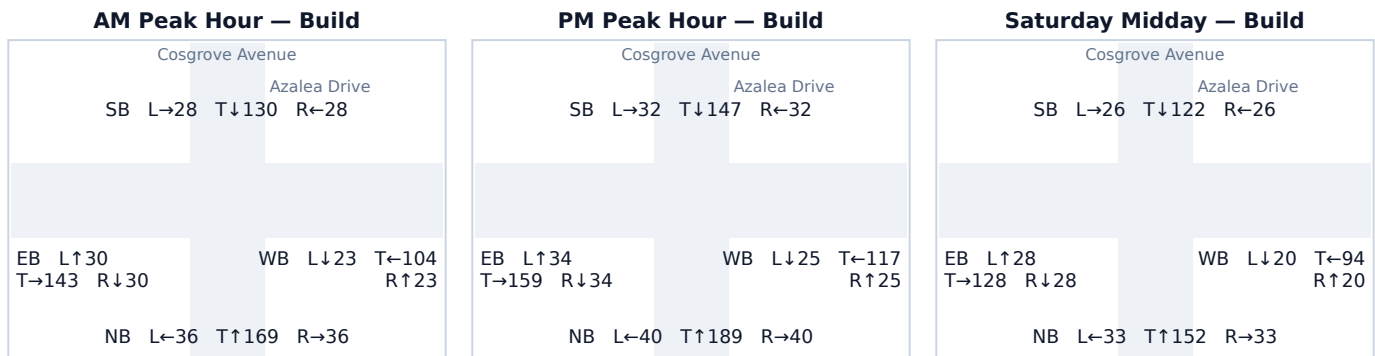
Movement	Project trips	% of project trips
NB Through	8	40%
SB Through	7	35%
NB Right	3	15%
WB Left	2	10%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.11 Cosgrove Avenue & Azalea Drive

Signal ID	charleston-sc-130
Location	Central Charleston (SC) · 0.67 mi from site
Added PM peak trips	15
Existing / No-Build (PM)	LOS B · 19.3 s/veh · v/c 0.48
Build (PM)	LOS B · 19.5 s/veh · v/c 0.49
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	265	4	269	0.33 → 0.33	17.0 → 17.1	B → B	180
SB	203	8	211	0.25 → 0.26	16.1 → 16.2	B → B	136
EB	224	3	227	0.28 → 0.28	16.4 → 16.4	B → B	147
WB	167	0	167	0.21 → 0.21	15.6 → 15.6	B → B	105

Affected movements (PM peak project trips)

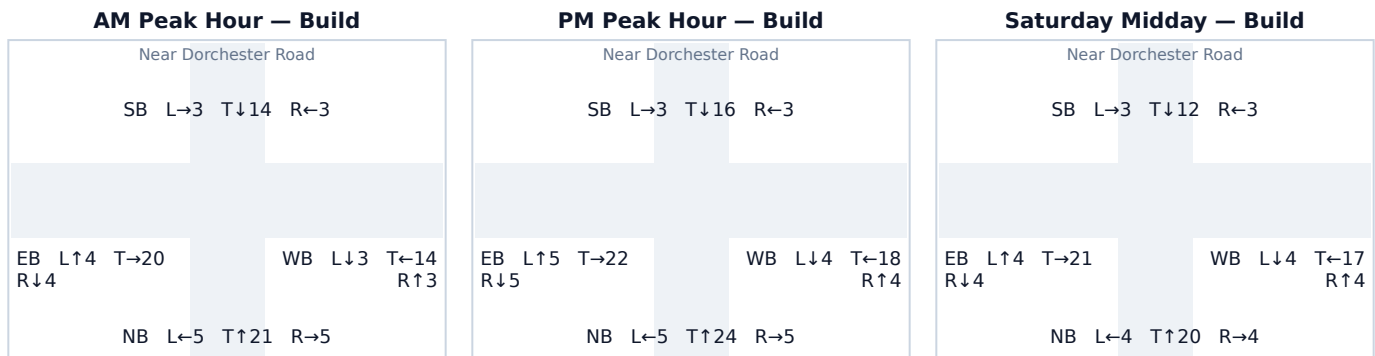
Movement	Project trips	% of project trips
SB Through	5	33%
NB Through	4	27%
SB Right	3	20%
EB Left	3	20%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.12 Near Dorchester Road

Signal ID	charleston-sc-743
Location	NW Charleston (SC) · 0.71 mi from site
Added PM peak trips	13
Existing / No-Build (PM)	LOS B · 14.1 s/veh · v/c 0.06
Build (PM)	LOS B · 14.2 s/veh · v/c 0.06
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	33	1	34	0.04 → 0.04	14.0 → 14.0	B → B	20
SB	22	0	22	0.03 → 0.03	13.8 → 13.8	B → B	13
EB	28	5	32	0.03 → 0.04	13.9 → 14.0	B → B	19
WB	19	7	26	0.02 → 0.03	13.8 → 13.9	B → B	15

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	6	46%
EB Through	5	38%
WB Left	1	8%
NB Right	1	8%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.